Semester II **B. Tech CSE (AI)**

BCSAI 207: COMPUTER AIDED GRAPHICS & DRAFTING LAB

0L + 0T + 4P + 2C **MM 100**

Introduction: Principles of drawing, conventional representation of machine components and materials, lines, types of lines, dimensioning types, rules of dimensioning, Computer aided drafting: Introduction to computer aided drafting, advantages and applications of CAD, concepts of computer aided drafting using AutoCAD, basic drawing and modify commands.
[Discussion on AutoCAD software and drawing in Sketch book]

Isometric Projections: Isometric projection of planes and solids.
[At least 3 Problems in AutoCAD and 3 problems in Sketch book]

Orthographic Projection: Introduction to orthographic projection, concept of first angle and third angle projection, drawing of simple machine elements in first angle projection.
[At LEAST 4 problems on AutoCAD and 4problems in sketch book]

Section of solids: Introduction to sectional views, Section of right solids by normal and inclined planes.
[At least 2 problems on AutoCAD and 2problems in sketch book]

Conventional representation of materials: Common features, Springs, Gear Assemblies, Materials, Interrupted views and Braking of Shaft, Pipe, Bar, Surface finishing & Machining Symbols
[At least 2 problems on AutoCAD and 2 problems in sketch book]

Miscellaneous: Welded joints, riveted joints, Belt and pulleys, screw fasteners, Bearings; Ball, roller, needle, foot step bearing Belt and pulleys, pipe joints.
[At least one problem from each on AutoCAD and sketch book preparation of all topics]

Text/Reference Books

1. N.D. Bhatt, Elementary Engg. Drawing, Chartor Pub. House, Anand, India.
2. D. N. Johle, Engineering Drawing, Tata Mcgraw-hill Publishing Co. Ltd..
3. P.S. Gill, Engineering Graphics.
4. N.D. Bhatt, Machine Drawing, Chartor Publishing house, Anand, India.
5. Warren J. Luzzader, Fundamentals of Engineering Drawing, Prentice Hall of India, New Delhi.
6. Fredderock E. Giesecke, Alva Mitchell & others, Principles of Engineering Graphics, Maxwell McMillan Publishing.